

EXPERIMENT-3- Playfair Cipher Algorithm

AIM: To write a C program to implement the Playfair Cipher encryption technique using a 5×5 matrix constructed with a keyword. The plaintext is encrypted two letters at a time using the Playfair encryption rules.

CODE:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main()
{
    char key[]="MONARCHY", p[100], m[5][5], used[26]={0};
    int i,j,k=0,r1,c1,r2,c2;

    for(i=0;key[i];i++)
    {
        char c=toupper(key[i]);
        if(c=='J') c='I';
        if(!used[c-'A'])
            m[k/5][k++%5]=c,used[c-'A']=1;
    }

    for(i=0;i<26;i++)
```

```
if(i!=9 && !used[i])
```

```
    m[k/5][k++%5]='A'+i;
```

```
printf("Enter plaintext: ");
```

```
scanf("%99s",p);
```

```
for(i=0;p[i];i+=2)
```

```
{
```

```
    char a=toupper(p[i]), b=p[i+1]?toupper(p[i+1]):'X';
```

```
    if(a=='J') a='I';
```

```
    if(b=='J') b='I';
```

```
    if(a==b) b='X';
```

```
for(j=0;j<25;j++)
```

```
{
```

```
    if(m[j/5][j%5]==a) r1=j/5,c1=j%5;
```

```
    if(m[j/5][j%5]==b) r2=j/5,c2=j%5;
```

```
}
```

```
if(r1==r2)
```

```
    printf("%c%c",m[r1][(c1+1)%5],m[r2][(c2+1)%5]);
```

```
else if(c1==c2)
```

```
    printf("%c%c",m[(r1+1)%5][c1],m[(r2+1)%5][c2]);
```

```
else
```

```
    printf("%c%c",m[r1][c2],m[r2][c1]);
```

```
}
```

```
return 0;
```

```
}
```

INPUT:WORLD

OUTPUT:

A screenshot of a terminal window. The title bar shows the number '48' and a green cursor icon. Below the title bar is a toolbar with icons for window management (minimize, maximize, close), a search icon, a settings icon, and a help icon. The terminal content shows the prompt 'Enter plaintext: WORLD' followed by the input 'NMTBZ'. Below this, the output is displayed in green text: '..Program finished with exit code 0' and 'Press ENTER to exit console.'